Helwan Univeristy
Faculty of Engineering
Elect., and Comm. Eng. Dept.

Optical Communication
3rd year (Comm.)

Sheet 1

1. In a step-index fiber, the relative refractive index difference is 2% and n_{clad} =1.4. Find

   a. The n_{core}.
   b. The critical angle.
   c. The N.A.
   d. The acceptance angle.

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2. An optical fiber is being designed. It must have an acceptance angle of 37.5°. Its cladding index is 1.38. Find its core index.

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3. The velocity of light in the core of a step index fiber is 2.01x10^8 m/sec, and the critical angle at the core-in cladding interfaces 80 degree. Determine the numerical aperture and the acceptance angle at the fibre in air, assuming it has a core diameter suitable for consideration by ray analysis. The velocity of light in vacuum is 2.998x 10^8 m/sec.

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4. A single mode step index fiber has a core diameter of 7 µm and a core refractive index of 1.49. Estimate the shortest wavelength of the light which allows single-mode operation when the relative refractive index difference for the fiber is 1%.

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5. In the previous problem, it is required to increase the fiber core diameter to 10 µm whilst maintaining the single-mode operation at the same wavelength. Estimate the maximum possible refractive index difference for the fiber.

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6. A single mode step index fiber which is designed for operation at wavelength of 1.3 µm has core and cladding refractive index of 1.447 and 1.442 respectively. When the core diameter is 7.2 µm, confirm that the fiber will permit single mode transmission and estimate the range of wavelengths over which this will occur.

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7. A single mode step index fiber has a core and cladding refractive indices of 1.498 and 1.495 respectively. Determine the core diameter required for the fiber to permit its operation over the wavelength range 1.48 and 1.60 µm. Calculate the new fiber core diameter for the fiber to enable single-mode transmission at a wavelength of 1.3 µm.

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8. Consider a multimode step-index fiber with a 62.5 µm core diameter and a core-cladding relative index difference of 1.5%. If the core refractive index is 1.480, estimate the normalized frequency of the fiber and the total number of modes supported in the fiber at a wavelength of 850 nm.

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9. Consider a multimode step-index optical fiber that has a core radius of 25 µm, a core index of 1.48, and an index difference=0.01. How many modes are in the fiber at wavelengths 860, 1310, and 1550 nm? Comment on the results.

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10. Consider three multimode step-index optical fibers each of which has a core index of 1.48 and an index difference= 0.01. Assume the three fibers have core diameters of 50, 62.5, and 100µm. How many modes are in these fibers at a wavelength of 1550 nm? Comment on the results.

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11. A certain single-mode step-index fiber has a Mode Field Diameter MFD = 11.2 µm and V = 2.25. What is the core diameter of this fiber?

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12. A step-index fiber has a radius a = 5 µm, core refractive index n_1 = 1.45, and fractional refractive-index change Δ = 0.002. Determine the shortest wavelength λ, for which the fiber is a single-mode waveguide. If the wavelength is changed to $\lambda c/2$, identify the indices (l, m) of all the linearly polarized modes.

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13. If a laser light with a wavelength λ=0.63 µm is incident into an optical fiber with a radius a=4.5 µm, that was originally designed to be a single-mode fiber (V = 2.0, @ λ= 1.55 µm, n_1 = 1.458) at the wavelength λ =1.55 µm, how many modes will the fiber support?

a- List the ($\nu\mu$) LP mode(s) in the fiber.
b- Calculate the numerical aperture NA of the fiber.
c- Conversely, if a laser light with wavelength λ= 1.55 µm is incident into an optical fiber that was originally designed to be a single-mode fiber (V = 2.0, @ λ= 0.63 µm, n_1=1.458) at wavelength λ= 0.63 µm, how many modes will the fiber support? What is (are) the value(s) of the propagation constant(s)?

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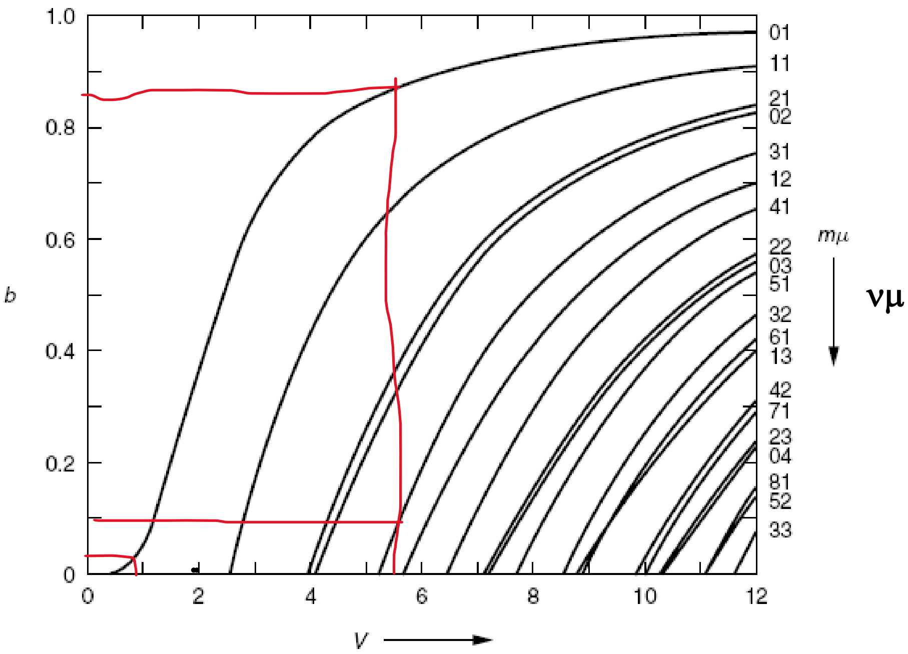
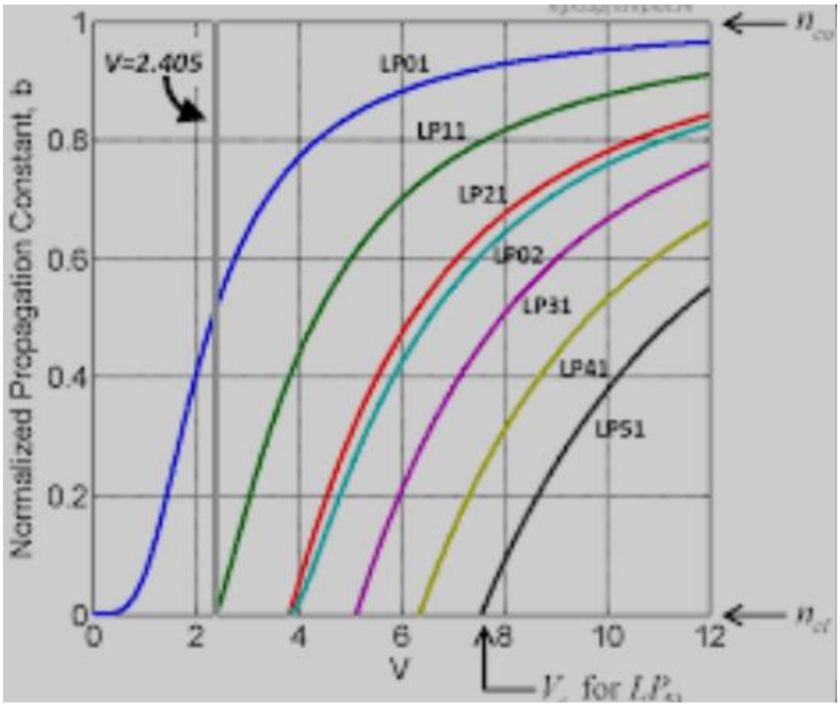
14. A step index SM fiber is designed such that the fundamental mode diameter is 10 microns at the wavelength of λ=1550 nm and a normalized effective index b = 0.4 using a cladding index of 1.5, calculate:

a. The fiber normalized frequency V.
b. The fiber core diameter.
c. The fiber numerical aperture.
d. The step index used.
e. Th effective index of the fundamental mode.
f. The number of modes in the fiber at 650 nm.
g. The fiber length at which the highest order mode and the lowest order modes will have a phase difference of π.


1.0
0.8
0.6
0.4
0.2
0
b
0
2
4
6
8
10
12
V
01
11
21
02
31
12
41
22
03
51
32
61
13
42
71
23
04
81
52
33
mμ
νμ


V=2.405
LP01
LP11
LP21
LP02
LP31
LP41
LP51
Normalized Propagation Constant, b
1
0.8
0.6
0.4
0.2
0
0
2
4
6
8
10
12
V